CodGen

Virtual Internship Program • Machine Learning

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Week 1 Task

Data Exploration & Preprocessing

Internship: Machine Learning | Duration: 1 Week | Level: Beginner–Intermediate

Objective

This week you will build a solid foundation by learning how to load, explore, clean, and visualise real-world datasets. These are the essential first steps of any Machine Learning project.

Tools Required

Tool	Purpose
Python 3.x	Core programming language
Pandas	Data loading & manipulation
NumPy	Numerical operations
Matplotlib	Basic data visualisation
Seaborn	Statistical visualisation
Jupyter Notebook	Interactive coding environment (or Google Colab)

Dataset

Choose **one** of the following datasets for this week's tasks:

Dataset	Problem Type	Source
Titanic	Binary classification — predict passenger survival	https://www.kaggle.com/datasets/titanic
Iris	Multi-class classification — predict flower species	https://archive.ics.uci.edu/dataset/53/iris

Tasks

Task 1 — Load & Explore the Dataset

- Load the dataset using `pd.read_csv()`
- Display the first 5 rows using `.head()`
- Check the shape: number of rows and columns
- Check data types using `.dtypes`
- Get basic statistics using `.describe()`
- Identify missing values using `.isnull().sum()`

Task 2 — Handle Missing Values

- Identify which columns have missing values
- For numerical columns: fill missing values with mean or median
- For categorical columns: fill with mode or drop the rows
- Justify your choice in a markdown cell

Task 3 — Encode Categorical Columns

- Identify categorical (text) columns in the dataset
- Apply Label Encoding using `sklearn.preprocessing.LabelEncoder`
- OR apply One-Hot Encoding using `pd.get_dummies()`
- Explain which method you used and why

Task 4 — Visualise the Data

- Plot a histogram for at least 2 numerical columns
- Plot a bar chart for at least 1 categorical column
- Create a correlation heatmap using Seaborn
- Add proper titles and labels to every plot

Task 5 — Write Your Observations

- After completing the above tasks, write 3 to 5 key observations
- What patterns or insights did you find in the data?
- Which features seem most important? Which seem irrelevant?
- Write these in a markdown cell at the end of your notebook

Submission Instructions

What to submit	Your complete Jupyter Notebook (.ipynb) — all cells must be run and outputs visible
Where to submit	https://forms.gle/k6ddCEN5KHYL1zxp8
Naming format	YourName_Week1_MLInternship.ipynb

Tips for success

- Use Google Colab if you don't have Jupyter installed locally
- Comment your code — explain what each step does
- If you encounter an error, try to debug it yourself first
- Quality of observations matters more than quantity of code